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Battery Management System Health Monitoring Using Artificial Intelligence

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Abstract—A battery-management system solves these typical limitations and improves the efficiency of battery modular handling. A battery state of health (SOH) examination is a vital step in preventing potential battery problems and ensuring safety. This paper presents a systematic literature analysis on battery health estimate methodologies in order to provide beneficial data to the scholarly field. Thus, the incorporation of innovations and emerging technologies allows for the implementation of continuous surveillance with an emphasis on Industry 4.0. Employing Artificial Intelligence technologies, the present research investigates several techniques and methods for calculating the SOH of Li-ion battery. The study demonstrated that AI approaches such as Artificial Neural Networks (ANN), Support Vector Machines (SVM), and Fuzzy Logic are capable of correctlyenvisaging the status of Li-ion batteries. At last, the paper makes important suggestions, like integrating edge computing with AI prototype-based products.

Keywords—AI, ANN, SVM, Fuzzy Logic, Battery, State of Health, Battery management system, State of Charge (SOC).

I. Introduction

Estimating battery state-of-health (SOH) is critical for addressing safety problems and maximising battery use. For estimate and regulate operations in an enhanced BMS, exact battery designs using a simpler computing method are preferred. Variability in SOH in systems of batteries are frequently dictated by operational conditions. Energy fading, capability, and resistance inside have traditionally been used to determine battery ageing levels. Knowing SOH is useful for prompt replacement of batteries and appropriate use to increase battery component cycle life. Furthermore, the cells' total resistance increases with a period of time limiting their maximum electrical production. Among of the possible deterioration processes that induce power fading and resistivity rising in batteries made from lead acid are positively grid rust, transpiration of water, design active substance decomposition, and separators metallization [1].

According to battery life standards and instructions, a battery should be changed as soon as it achieves less than 80% of its original capacity rating [2]. In other cases, yet, a battery gets unstable earlier owing to increasing resistance within, resulting in power withering. Considering all of these factors into consideration, it is plausible to conclude that the battery's charge-discharge curves, open circuit voltage (OCV) curves [3], and other cell metrics that fluctuate significantly with battery age may foresee battery death in preparation. Though SOH shows battery recital deviation, it cannot be straight determined. Notwithstanding the fact that many technologies and methodologies are used to thoroughly analyse battery SOH, determining health-related battery characteristics is a difficult process.

Among the recent developments has been the employment of significant artificial intelligence technologies to improve estimation performance. Yet, the enormous need of information as well as the expensive nature of computer facilities remain issues to be addressed. However, a more fair and comprehensive strategy is required to increase SOH estimation performance. This comprehensive strategy takes into account several estimating approaches as well as modern embedded sensor technologies. Previous evaluations mostly concentrated on categorizing, developing, and summarising various types of SOH estimating strategies based on algorithms, accuracy, and difficulties.

With this motivation, the involvement of study as tracks:

• This review study not only addresses SOH approaches in depth with separate classification, but it also incorporates improved data generating strategies in addition to novel factor AI modules.

• The paper also gives insight on potential remedies for data scarcity, incomplete number of cycles difficulties, and multi-time scale concerns.

• This open-ended conversation outlines several prospective study areas for various estimating methodologies that remain unexplored.

This study structured in the manner: Chapter 2 grants the battery management system (BMS) with help of AI Chapters 3 talks about ANN in BMShealth, Chapter 4 thoroughly converse Fuzzy Logic in states of health techniques, Chapter 5 talks about SVM in BMS health and thesuggestions of SOH are conversed in Chapter 6. Lastly, Chapter 7covers the conclusions.

II. Battery Management System

A BMS is a computerized device that assesses the temperature, voltage, and current of a battery's cells in actual time and records data for cell balance, thermal control, protection of cells (over-current, over-voltage, and over-heat), SOH, and SOC calculations. Figure 1 shows a structure depicting the elements that impact the SOH and SOC of lithium-ion batteries, as well as their BMS operations. A BMS collects data on electrical current, voltage, and temperature. Normalizing, prototype and organisation of data results in state estimate in the form of SOC and SOH employing a timer, storage modules, and a user interface [5]. It also protects from over voltages, under voltages, and overheating. Figure 1 depicts all of the previously stated BMS functionalities.

In EVs, the BMS may include a range of motors, controllers, and sensors. BMSs are in charge of battery safety, maintaining packs within permissible electrical current, voltage, and thermal variables, and properly measuring parameters related to batteries [6].

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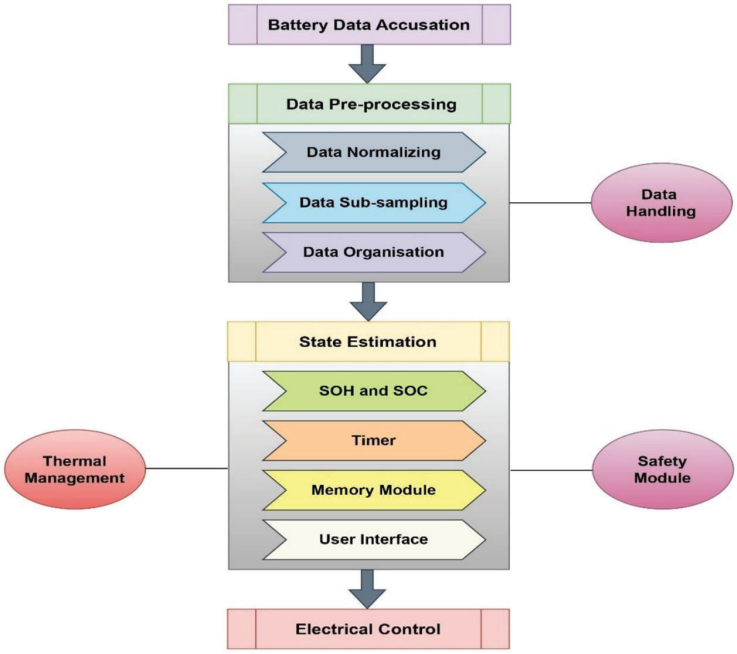

Fig. 2. Framework of the factors influencing battery SOH with its functions

In regard to equipment framework, three fundamental morphologies have been employed: modularity architectures, centralised architectures, and dispersed architectures. These ideas might be utilized to provide a wide foundation having basic capabilities. At the tracking layer, data is collected by several sensors situated inside the battery package. The BMS is connected to all elements of the battery system in addition to the engine control processors. BMSs have an extensive record of employing safety [7]. The proposed BMS, contrary to popular belief, employs a greater number of sensors than current safety circuit concepts, enabling innovations such as the inclusion of accurate alerts and controllers to minimise overcharge, over-discharge, and overheating.

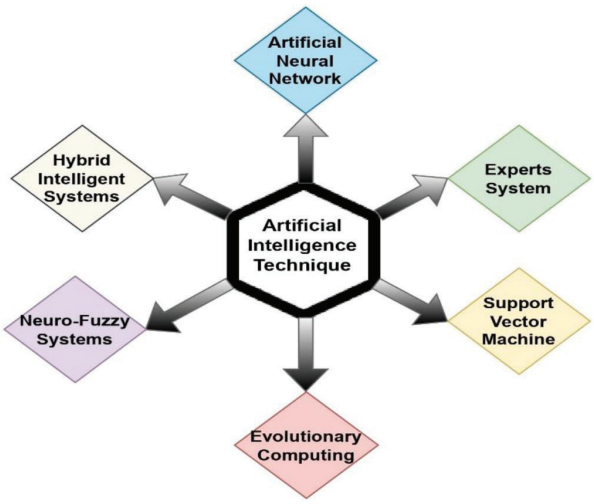

Fig. 3. AI Techniques in BMS

There are various conventional strategies, but AI-based solutions for boosting variable reactivity and reliability have gained prominence. BMSs are at present implemented with a present control mechanism that is not adjusted or modified over the system's lifespan [8]. The battery packs, on the other hand, have a pretty long longevity, with various guarantees spanning from 5 to 10 years. Due of the system's lengthy lifespan, the operating climate may change. To increase the characteristics of performance, these improvements might require modifications to the monitoring and control approach. The importance of SOH and SOC for the BMS goes above the precision of its computations [9]. A BMS is in charge of not only conveying battery data to the consumer, as well as of safeguarding the battery. Figure 2. below depicts the many AI strategies that may be utilised for state estimate.

III. ANN In BMS

The ANN is composed of hubs or neurons that are similar to the neurons in humans. A neural network (NN) is essentially a multi-layer architecture with inputs, results, and hidden layers that are comprised of neurons with requirements of the system. It is critical to precisely assess the battery's level of health so as to improve the efficiency of a batteries-powered device. When the level of charge of the Li-ion battery is 100%, the AC resistivity of the power source at all frequencies may be measured using electrochemical impedance spectroscopy (EIS) and utilised as training information for the neural network. Additional ageing battery corroborated the data, which reveals that when using the top performing neural network, the highest possible absolute error of the predicted SOH is only 1.31% [10].

The brand-new on-line synthesis technique that predicts SOH and residual usable lifespan under continuous electrical discharging by combining partial incremental power and ANN. The initially generated IC curves are calculated, and then these IC arcs are smoothed using filter algorithms.Then, through the partial IC curves, various characteristics are retrieved. The feature variables with the greatest coefficient of correlation are chosen to be the input for ANN models after the relationship among these characteristics values as well as capacity has been examined. The suggested technique has good accuracy as well as dependability for SOH and RUL

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calculation, according to the instruction and validation findings [11].

Using actual data taken from the vehicle during its everyday excursions, a nonlinear autoregressive with external input (NARX) artificial neural network may be used to calculate the SOC of an EV. Voltage, electrical current, and four distinct battery pack degrees are used as inputs, while SOC is used as an output, to train the network's behavior. The NARX-ANN had been trained off-line, and after testing with fresh data sets, it consistently produced mean squared errors less than 1e-6 [12].

An ANN-based capacity predictor is created to forecast the obtainable number of cells for batteries after this connection is learned. To get a more precise minimal resistance for the ANN-based capacities prototype an enhanced measuring precision online voltage measuring approach is proposed. In order to determine the SOC parameters for the cells in the battery, a SOC rebalancing controller receives the projected capability from the ANN load estimator. Whenever SOH or capacity estimate capability is turned on, the SOC value of a cell in a battery with a lower health rating is reduced because it has less capacity accessible than a battery cell with higher health. The framework draws power coming from the battery cells that have less SOH at a slower pace and relies power from the battery prototype(s) that have greater SOH at a greater pace as a result of the aforementioned mechanism and the management concept of the provided SOC balancing administrator, causing all battery cells to come to their end of release simultaneously. As a consequence, the system's operating duration is increased, each battery cell receives the maximum amount of power, and over discharging battery cells containing less SOH is prevented [13].

IV. Fuzzy Logic System In BMS

Usuallytalking, fuzzy logic controllers (FLSs) are inadequate for calculating the battery prototype's SOC and SOH. The SOC and SOH of the battery pack were determined using a fuzzy technique by only one study team. Prior till this, most studies on adaptable battery control have only used fuzzy logic or neural network models. Yet, study has started to lean towards integrating these two fields in

recent years. Neuro-fuzzy logic is now being studied and used more often due to the capacity to balance off one limitation with another's limitations [14].

Fuzzy logic and neural networks are used to lessen the need for data from experiments. Neural networks and fuzzy logic offer adaptive answers to the bulk of BMS problems as a substitute to static ones. The application of deep neural networks in BMSs is constrained by the absence of an information set that can educate the system about the operational conditions at the point of system installation [15]. On the contrary, such information may be continuously and instantly gathered.

The objective of the neuro-fuzzy technique is to modify the fuzzy system's properties by means of a process of learning utilising input-output data used for training. The ANFIS framework uses Sugeno fuzzy rules. There are five of them. The input parameters make up layer 1 of the stack. The membership functions layer in Layer 2 controls the weights of the participation function. In order to describe fuzzy groupings of related variables that are input, this layer sets the values for membership and receives input information from the top layer. The components from layer one is contained in layer three, sometimes referred to as the rule layer. Each rule's level of action is set by this layer. The defuzing layer is the fourth layer. The defining layer offers the resultant values generated through rule derivation. The output layer, Layer 5, collects each of the signals from the layers before it and converts the outcome of the fuzzy categorization into a distinct value [16].

A comprehensive study of the SOH estimate of Li-ion batteries beneath varying operating situations was provided by Venugopal et al. [17]. The evaluation's methodology relied on an in-depth categorization of a range of conventional and cutting-edge SOH estimating techniques, including direct approaches, model-based techniques, adaptive-filter techniques, and data-computational techniques. The benefits and drawbacks of direct, model-based (electro-chemical, equivalent circuit model), and adaptable filters. and data computing techniques (such as SVM, ANN, and fuzzy logic) were also included in the paper. Figure 3. displays a usual fuzzy logic controller block layout. The fuzzifier's job is to transform sharp numbers into fuzzy values, and the fuzzy rule-base holds information about how the procedure works. The resultant parameters are acquired using the defuzzifier, which transforms values that are fuzzy to crisp values, while the function of membership specifies the value of the input rule-base.

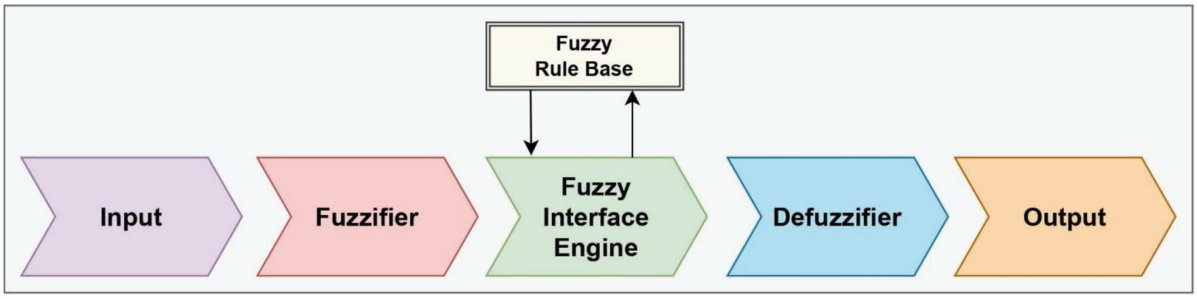

Fig. 4. Fuzzy logic controller

V. Support Vector Machine In Bms

Techniques for SOC estimation based on SVM were described in [18]. As an illustration, [18] estimated the battery's SOC as an expression of power and flow. On the contrary hand, metrics, the current, voltage, and temperatures were used as sources for training the model using SVM and calculate the SOC. It has been shown that this theory outperforms the usual model based on empirical risk reduction employed by traditional neural networks. The learning approach of SVMs is used to resolve an issue with quadratic

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programming that has a global optimum answer; though neural network models could face problems with local reduction, SVM may generalise more effectively as a consequence. The idea that SVMs are meant to entirely replace ANN is not instantly obvious.

To determine the SOH statistics of ability, capability, and opposition, a two-step method of simulator training and simulated testing has been used to battery data acquired with a common EV present characteristic [19]. During the first stage, an SVM undergoes training utilising current, temperatures, and SOC as parameters and voltages as the result. The generated model uses the true current/temperature SOC characteristic from the pertinent real conventional test as a voltage look-up table for speculative prototype/temperature SOC input. The results of the simulated test can subsequently be used to compute impedance and capacity, just like in actual conventional testing.

The study [20] suggests a PSO-SVM-based technique for online SOC-SOH calculation. A relationship with the battery's health condition and its level of charge is discovered by examining the NASA-provided public batteries condition data. To be more precise, the relationship between the slope of the battery's SOC as well as time increases as the battery ages, whereas the slope of the battery's discharging voltage and duration tends to rise. In order to accurately predict the battery's health state, the research [20] uses the battery's SOC variation frequency and discharging voltage shift rate as defining variables of SOH and optimises the SVM utilizing the method of particle swarm technique. Last but not least, the algorithm's viability and suitability for use in actual manufacturing and daily life has been demonstrated through algorithm evaluation.

TABLE II. COMPARATIVE ANALYSIS ESTIMATING BATTERY SOH

Methods	Advantages	Disadvantages	Research Gaps
ANN [22]	• Online computation of battery is possible. • Taken care of non-linearity.	• Huge data is required. • Required proper management of battery degradation modelling.	Outer aspects i.e., climate change or the charging used in batteries can be taken for better results.
SVM [23]	• Online computation of battery is possible. • Taken care of non-linearity.	• Outer factors like temperature or the errors will be considered during SOH of battery.	Sampling errors also be taken to solve non-linear systems.
Fuzzy Logic [24]	• Higher reliability.	• Computational unit is required. • Large datasets required for high accuracy	More changes of accuracy if outer/ internal factors considered.

VI. SUGGESTIONS

- Batteries made with lithium ion are a significant electrochemistry advancement, and applications such as ESS have increased significantly [21]. Li-ion batteries have made a name for their products in the household electronics and electric vehicle industries without facing any competition. By making ingenious changes to the many useful elements of the BMS, they are going to dominate the automobile mobility and green energy storage industries.
- By employing Machine Learning (ML) and Neural Networks, one may capitalise on the untapped possibilities and prospect of battery life cycling management. Integrating contemporary electronics with information from testing and training can improve the performance of the batteries used in EVs. The application of prediction awareness, which is generated from empirical data collected in laboratories, may be utilized to estimate battery life, identify issues and reasons for them, and prevent mistakes from happening.
- High-accuracy SOH estimations may be obtained by creating altered IC curves using similar circuit simulations and the least squares optimisation technique. Once more, the validation procedure needs to take into account a number of variables, including different levels of discharge, states of charge, and temperatures.
- The developed methods and methodologies should not be restricted to the level of a single cell; rather, the corresponding research should advance to the level of a battery module or system. Although controlling the variability in cell ageing is challenging, all of the aforementioned elements must be taken into account to develop an appropriate method for SOH assessment.

VII. CONCLUSION

In this study effort, the battery management system's primary parameter condition of health had been covered in detail. Finding the finest AI technology for BMS is the main aim of this endeavour. As a result, a thorough literature assessment of the selected AI technologies is carried out, emphasising the difficulties faced. Due to its superior productivity, longer duration, lower discharge prices, and higher power and energy concentrations, lithium-ion batteries were the subject of this suggested study. In order to guarantee safe and dependable battery operation in electric cars, the main parameters and their functions in a BMS's secure functioning and functioning cycle have been detailed. In order to produce a dependable, compact, and ecological battery pack in the not-too-distant future with a longer life, this article can succinctly inform scientists and manufacturers about the present and emerging topics that are being studied in the discipline of batteries.

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